ANSWERS

1. a) $(A+B)(A+\overline{B})(\overline{A}+C) = (AA+A\,\overline{B}+BA+B\,\overline{B})(\overline{A}+C)$ (distributivity)

$= (A+A\,\overline{B}+BA+B\,\overline{B})(\overline{A}+C)$ (idempotence)

$= (A+A\,\overline{B}+BA+0)(\overline{A}+C)$ (complement)

$= (A+A\,\overline{B}+BA)(\overline{A}+C)$ (identity)

$= A(\overline{A}+C)$ (Rule 10 - or distributivity, etc.)

$= A\overline{A}+AC$ (distributivity)

$= 0+AC$ (complement)

$= AC$ (identity)

1 mark for answer in simple form. 3 marks for method (which may vary).

b) K-map in Fig 1.1 gives $Y = X_2\overline{X_0}+X_2X_1+\overline{X_2}X_0$.

1 mark for 1s in right cells. 2 marks for minimal groups identified. 1 mark for final expression.

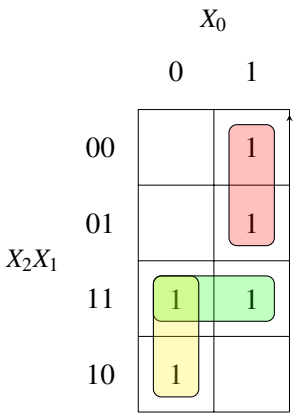


Figure 1.1: K-map

c) If $B = 1$, $C = 0$, and A transitions, then a glitch might be observed. Modified SOP $F = AB+\overline{A}\,\overline{C}+B\overline{C}$

2 marks for identifying constant inputs. 1 mark for final SOP.

2. a) No, because it requires $B[n]$ to be responsive to $A[n]$ (i.e. in the same cycle).

2 marks for any reasonably justified correct answer.

b) We need to record the values of $A[n-1]$ and $A[n-2]$ to be able to compute this function. Hence we need 4 states minimum.

3 marks for any reasonably justified correct answer.

c) The diagram is shown in Fig. 2.1.

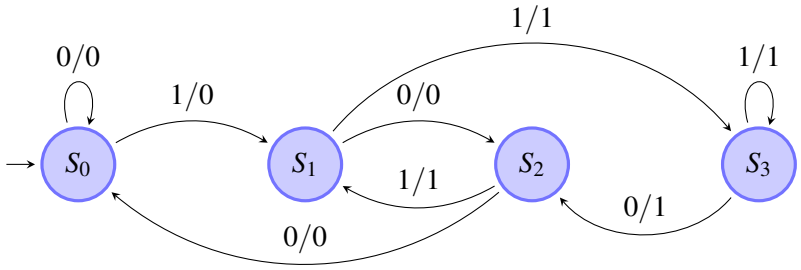


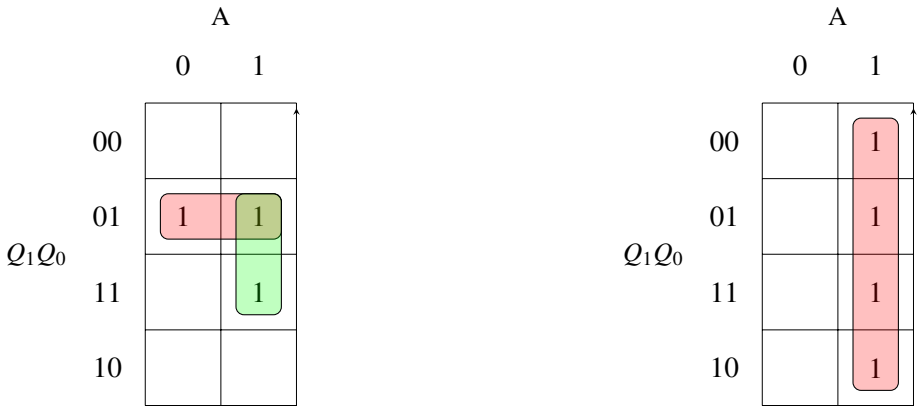
Figure 2.1: Mealy model diagram

1 mark per correct edge and associated label.

d) A natural encoding is $S_0 \mapsto 00$, $S_1 \mapsto 01$, $S_2 \mapsto 10$, $S_3 \mapsto 11$

2 marks for any encoding that utilises a distinct binary vector per state.

e) The K-maps are in Fig. 2.2, with the corresponding SOPs in the subcaptions.



(a) $Q_1^+ = \overline{Q_1}Q_0 + Q_0A$ (b) $Q_0^+ = A$

Figure 2.2: Next state K-maps

For each of the next state bits: 2 marks for a correct identification of minterms, 2 marks for a correct identification of groups, 2 marks for a correct minimal SOP.

f) The K-map is in Fig. 2.3, with the corresponding SOP in the caption.

1 mark for a correct identification of minterms, 1 mark for a correct identification of groups, 1 mark for a correct minimal SOP.

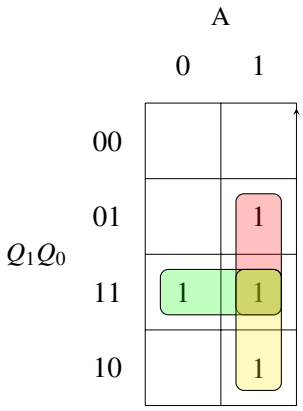


Figure 2.3: Output K-map $B = Q_0A + Q_1Q_0 + Q_1A$

g) 1 mark per row

		Comments (not required in answer)
R0	0xFF9C	(Negative immediate numbers are sign extended)
R1	0x0000	
N,Z	1,0	
C,V	1,0	

h)

R6(i) = R4(i-6) // 0 if i < 6 (1 mark)

R5(i) = R4(i+10) // 0 if i > 21 (1 mark)

Also allow: (R6:R5)(i+10) = R4(i) // for $0 \leq i \leq 31$

u(R6:R5) = $(2^{10} - 1)$u(R4) = 1023u(R4) (4 marks)

Allow 2 marks for u(R6:R5) = 1024u(R4) before subtraction.

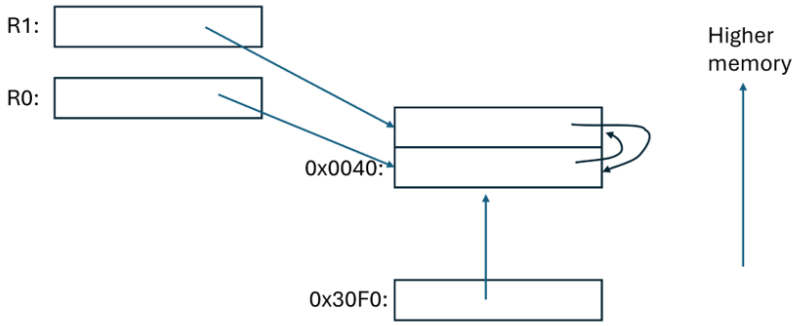
i)

Address	Data
0x0041	0x0040
0x0042	0x003F
0x30F0	0x0040

1 mark each correct address, 1 mark for the corresponding data item (6 marks total)

j) 4 marks diagram:

1 mark: correct registers shown + pointers from register to memory
1 mark: correct memory addresses shown 0x40/0x41
1 mark correct memory address: 0x30F0
1 mark: correct in memory pointer relationships / arrows



3.

(a) $T1_{ISR}$ (longest interrupt) + $T4_{ISR}$ (higher priority interrupt) + T_{PRE} = 190

(4 marks method, 2 answer)

(b) $T1_{PRE} + T4_{ISR} + T_{PRE} = 10 + 20 + 10 = 40$

(as before but instead of all T1 we have only $T1_{PRE}$)

(4 marks method, 2 answer)

(c) No positive overflow: $(2^{15}-1) \geq N(2^5-1)$
No negative overflow: $2^{15} \geq N2^5$
Negative overflow is more severe constraint =>
$N \leq 2^{10} = 1024$ (3 marks)

Generalising: $N \leq 2^{16-n}$ (3 marks)

(d)

Entry:

```
ADD R7, #3
STR R0, [R7]
STR R1, [R7,#-1]
STR R2, [R7, #-2]
SETI
```

3 marks

Exit:

```
CLRI
LDR R0, [R7]
LDR R1, [R7,#-1]
LDR R2, [R7, #-2]
RETINT
```

3 marks

(e)

For correct operation no item on the stack must be overwritten. Therefore the SP update instruction must precede all of the store instructions. (4 marks). The answer above to d) is an example of this. (2 marks).